The Journal of Open Source Software

DeepInverse: A Python package for solving imaging inverse problems with deep learning

Julián Tachella[1*¶], **Matthieu Terris**[2*], **Samuel Hurault**[3*], **Andrew Wang**[4*], **Dongdong Chen**[5], **Minh-Hai Nguyen**[8], **Maxime Song**[14], **Thomas Davies**[4,5], **Leo Davy**[1], **Jonathan Dong**[7], **Paul Escande**[9], **Johannes Hertrich**[15], **Zhiyuan Hu**[7], **Tobías I. Liaudat**[13], **Nils Laurent**[6], **Brett Levac**[12], **Mathurin Massias**[10], **Thomas Moreau**[2], **Thibaut Modrzyk**[11], **Brayan Monroy**[6], **Sebastian Neumayer**[16], **Jérémy Scanvic**[1], **Florian Sarron**[8], **Victor Sechaud**[1], **Georg Schramm**[17], **Chao Tang**[4,5], **Romain Vo**[1], **and Pierre Weiss**[8]

1 CNRS, ENS de Lyon, Univ Lyon, Lyon, France **2** Université Paris-Saclay, Inria, CEA, Palaiseau, France **3** CNRS, ENS Paris, PSL, Paris, France **4** University of Edinburgh, Edinburgh, UK **5** Heriot-Watt University, Edinburgh, UK **6** Universidad Industrial de Santander, Bucaramanga, Colombia **7** EPFL, Lausanne, Switzerland **8** IRIT, CBI, CNRS, Université de Toulouse, Toulouse, France. **9** IMT, CNRS, Université de Toulouse, Toulouse, France **10** Inria, ENS de Lyon, Univ Lyon, Lyon, France **11** INSA de Lyon, Lyon, France **12** University of Texas at Austin, Austin, USA **13** IRFU, CEA, Université Paris-Saclay, Gif-sur-Yvette, France **14** CNRS UAR 851, Université Paris-Saclay Orsay, France **15** Université Paris Dauphine - PSL, Paris, France **16** Chemnitz University of Technology, Chemnitz, Germany **17** Department of Imaging and Pathology, KU Leuven, Leuven, Belgium ¶ Corresponding author * These authors contributed equally.

DOI: 10.xxxxxx/draft

Software

- Review
- Repository
- Archive

Editor:

Submitted: 26 May 2025
Published: unpublished

License
Authors of papers retain copyright and release the work under a Creative Commons Attribution 4.0 International License (CC BY 4.0).

DRAFT

Summary

DeepInverse is an open-source PyTorch-based library for imaging inverse problems. The library covers all crucial steps in image reconstruction from the efficient implementation of forward operators (e.g., optics, MRI, tomography), to the definition and resolution of variational problems and the design and training of advanced neural network architectures.

Statement of Need

Deep neural networks have become ubiquitous in various imaging inverse problems, from computational photography to astronomical and medical imaging. Despite the ever-increasing research effort in the field, most learning-based algorithms are built from scratch, are hard to generalize beyond their specific training setting, and the reported results are often hard to reproduce. `deepinv` overcomes these limitations by providing a unified framework for defining imaging operators and solvers. It leverages the popular PyTorch deep learning library (Paszke et al., 2019), making most modules compatible with auto-differentiation. The target audience of this library is both researchers in inverse problems (experts in optimization, machine learning, etc.) and practitioners (biologists, physicists, etc.). `deepinv` has the following objectives:

1. **Accelerate research** by enabling efficient testing, deployment and transfer of new ideas across imaging domains;
2. Enlarge the **adoption of deep learning in inverse problems** by lowering the entrance bar to new researchers and practitioners;
3. Enhance **research reproducibility** via a common framework for imaging operators, reconstruction methods, datasets, and metrics for inverse problems.

While other Python computational imaging libraries exist, to the best of our knowledge, deepinv is the only one with a strong focus on learning-based methods, providing a larger set of realistic imaging operators. SCICO (Balke et al., 2022) and Pyxu (Simeoni et al., 2024) are Python libraries whose main focus is variational optimization and/or plug-and-play reconstruction methods. These libraries do not provide specific tools for training reconstruction models such as trainers and custom loss functions, and do not cover non optimization-based solvers including diffusion methods, adversarial methods or unrolling networks. CUQIpy (Riis et al., 2024) is a library focusing on Bayesian uncertainty quantification methods for inverse problems. TIGRE (Biguri et al., 2025), ODL (Adler et al., 2018) and CIL (Jørgensen et al., 2021) mostly focus on computed tomography and do not cover deep learning pipelines for inverse solvers. There are also multiple libraries focusing on specific inverse problems: ASTRA (Van Aarle et al., 2016) and the related pytomography (Polson et al., 2025) define advanced tomography operators, sigpy (Ong & Lustig, 2019) provides magnetic resonance imaging (MRI) operators without deep learning, and PyLops (Ravasi & Vasconcelos, 2019) provides a linear operator class and many built-in linear operators. These operator-specific libraries can be used together with deepinv as long as they are compatible with PyTorch (for example, we provide a wrapper for ASTRA). Advanced libraries for inverse problems also exist in other programming languages such as MATLAB, including GlobalBiolm (Soubies et al., 2019) or IR Tools (Gazzola et al., 2019), but they are restricted to handcrafted reconstruction methods without automatic differentiation.

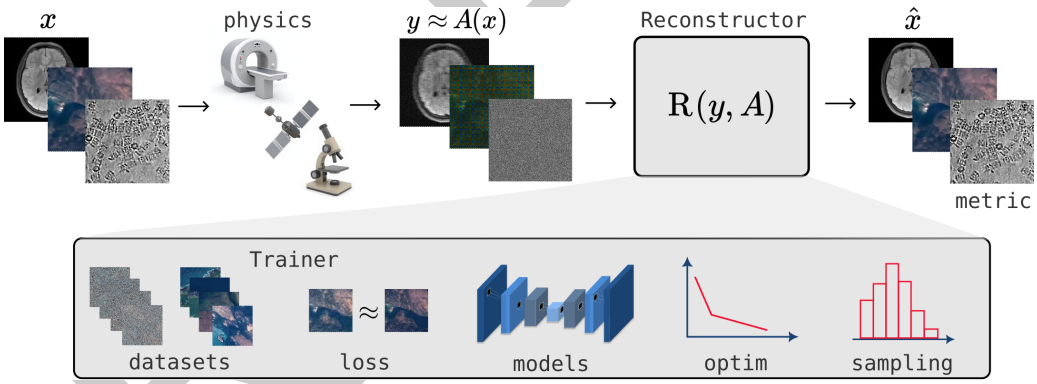

Figure 1: Schematic of the library.

Inverse Problems

Imaging inverse problems can be expressed as

$$y = N_\sigma(A_\xi(x)) \tag{1}$$

where $x \in \mathcal{X}$ is an image, $y \in \mathcal{Y}$ are the measurements, $A_\xi \colon \mathcal{X} \mapsto \mathcal{Y}$ is a deterministic (linear or non-linear) operator capturing the physics of the acquisition and $N_\sigma \colon \mathcal{Y} \mapsto \mathcal{Y}$ is a mapping that characterizes the noise affecting the measurements parameterized by σ (e.g. the noise level and/or gain). The forward operation is simply written in deepinv as `x = physics(y, **params)` where `params` is a dictionary with optional parameters ξ. This framework unifies the wide variety of forward operators across various domains. Most forward operators in deepinv are matrix-free, scaling gracefully to large image sizes. The library provides high-level operator definitions which are associated with specific imaging applications (MRI, computed tomography, radioastronomy, etc.), and allows users to perform operator algebra, like summing, concatenating or stacking operators. deepinv comes with multiple useful tools for handling linear operators, such as adjoint, pseudoinverses and proximal operators (leveraging the singular value decomposition when possible), matrix-free linear solvers (Hestenes et al., 1952) (Paige & Saunders, 1975) (Van der Vorst, 1992), and operator norm and condition number estimators

(Paige & Saunders, 1982). Many common noise distributions are included in the library such as Gaussian, Poisson, mixed Poisson-Gaussian, uniform and gamma noise. The table below summarizes the available forward operators at the time of writing, which are constantly being expanded and improved upon by the community:

Family	**Operators** A_ξ	**Generators** ξ
Pixelwise	Denoising, inpainting, demosaicing, decolorize	Mask generators, noise level generators
Blur & Super-Resolution	Blur, space-varying blur, downsampling	Motion, Gaussian, diffraction, product-convolution, and confocal blurs. Sinc, bilinear and bicubic antialias filters.
Magnetic Resonance Imaging (MRI)	Single and multi-coil, dynamic and sequential (All support 3D MRI)	Gaussian, random and equispaced masks
Tomography	2D parallel beam, 2D fanbeam, 3D parallel beam and 3D conebeam	
Remote Sensing & Multispectral	Pansharpening, hyperspectral unmixing, compressive spectral imaging	
Compressive	Compressed sensing and single-pixel camera	
Radio Interferometric Imaging	Monochromatic intensity imaging with narrow field-of-view	
Single-photon Lidar	TCSPC lidar	
Dehazing	Parametric haze model	
Phase Retrieval	Random operators and ptychography	Probe generation

Operator Parameterization

Most physics operators are parameterized by a vector ξ, which has a different meaning depending on the context. For instance, it can represent the projection angles in tomography, the blur kernel in image deblurring, the acceleration masks in MRI, etc. Integrating this parameter allows for advanced computational imaging problems, including calibration of the system (measuring ξ from y), blind inverse problems (recovering ξ and x from y) (Debarnot & Weiss, 2024) (Chung et al., 2023), co-design (optimizing ξ and possibly the reconstruction algorithm jointly) (Lazarus et al., 2019) (Nehme et al., 2020), robust neural network training (Gossard & Weiss, 2024) (Terris & Moreau, 2023) (Terris et al., 2025). To the best of our knowledge, this feature is distinctive and becoming essential in recent advances in image reconstruction.

DRAFT

Reconstruction Methods

The library provides multiple solvers that depend on the forward operator A_ξ and the noise distribution via σ. Our framework unifies the wide variety of solvers that are commonly used in the current literature:

$$\hat{x} = \mathrm{R}_\theta(y, A_\xi, \sigma) \tag{2}$$

where R_θ is a reconstruction network/algorithm with (optional) trainable parameters θ (see *"Training"*). In `deepinv` code, a reconstructor is simply evaluated as `x_hat = model(y, physics)`. The library covers a wide variety of approaches for building R_θ, which can be roughly divided into optimization-based methods, sampling-based methods, and non-iterative methods.

Optimization-Based Methods

These methods consist of solving an optimization problem (Chambolle & Pock, 2016)

$$\mathrm{R}_\theta(y, A_\xi, \sigma) \in \operatorname{argmin}_x f_\sigma(y, A_\xi(x)) + g(x) \tag{3}$$

where $f_\sigma \colon \mathcal{Y} \times \mathcal{Y} \mapsto \mathbb{R}$ is the data fidelity term, which can incorporate knowledge about the noise parameters σ, and $g \colon \mathcal{X} \mapsto \mathbb{R}$ is a regularizer that promotes plausible reconstructions. The `optim` module includes classical fidelity terms (e.g., ℓ_1, ℓ_2, Poisson log-likelihood) and offers a wide range of regularization priors:

Explicit Priors: The library implements several traditional regularizers, such as sparsity (Candès & Wakin, 2008), total variation (Rudin et al., 1992), wavelets (Mallat, 1999), patch-based likelihoods (Zoran & Weiss, 2011a), and mixed-norm regularizations (Kowalski, 2009). We also provide learned regularizers such as EPLL (Zoran & Weiss, 2011b) and PatchNR (Altekrüger et al., 2023).

Denoising Plug-and-Play Priors: Plug-and-Play (PnP) methods replace the proximal operator (Venkatakrishnan et al., 2013) or the gradient (Romano et al., 2017) of the regularizer g with a pretrained denoiser, often based on deep learning. The library provides access to widely used pretrained denoisers D_σ, many of them trained on multiple noise levels σ, including DnCNN (K. Zhang et al., 2017), DRUNet (K. Zhang et al., 2021), and recent diffusion-based denoisers such as DDPM (Ho et al., 2020) and NCSN (Song et al., 2020).

The `optim` module also includes solvers for the minimization problem in (3) using:

Optimization Algorithms: The library contains several classical algorithms (Dossal et al., 2024) for minimizing the sum of two functions, including proximal gradient descent, FISTA, ADMM, Douglas-Rachford Splitting, and primal-dual methods.

Unfolded Networks and Deep Equilibrium Models: Unfolded networks (Gregor & LeCun, 2010) are obtained by unrolling a fixed number of iterations of an optimization algorithm and training the parameters end-to-end, including both optimization hyperparameters and deep regularization priors. Standard unfolded methods train via backpropagation through the optimization algorithm, while deep equilibrium methods (Bai et al., 2019) implicitly differentiate the fixed point of the algorithm.

Sampling-Based Methods

Reconstruction methods can also be defined via ordinary or stochastic differential equations, generally as a Markov chain defined by

$$x_{t+1} \sim p(x_{t+1} | x_t, y, \mathrm{R}_\theta, A_\xi, \sigma) \text{ for } t = 0, \dots, T-1 \tag{4}$$

such that x_T is approximately sampled from the posterior $p(x|y)$, and R_θ is a (potentially learned) denoiser. Sampling multiple plausible reconstructions enables uncertainty estimates

DRAFT

by computing statistics across the samples. The sampling module provides implementations of the following methods:

Diffusion Models: In a similar fashion to PnP methods, diffusion models (Chung et al., 2022) (Kawar et al., 2022) (Zhu et al., 2023) incorporate prior information via a pretrained denoiser, however, they are linked to a stochastic differential equation (SDE) or an ordinary differential equation (ODE), instead of the optimization of (3).

Langevin-Type Algorithms: The library provides popular high-dimensional Markov Chain Monte Carlo (MCMC) methods such as Unadjusted Langevin Algorithm (ULA) and some of its variants (Laumont et al., 2022) (Pereyra et al., 2020), which define a Markov chain with stationary distribution close to the posterior distribution $p(x|y) \propto \exp(f_\sigma(y, A_\xi(x)) + g(x))$.

Non-Iterative Methods

Non-iterative methods are part of the `models` module, and include artifact removal, unconditional and conditional generative networks, and foundation models.

Artifact Removal: The simplest way of incorporating the forward operator into a network architecture is to backproject the measurements to the image domain and apply a denoiser (image-to-image) architecture D_σ such as a UNet (Jin et al., 2017). These architectures can be thus written as

$$\mathrm{R}_\theta(y, A_\xi, \sigma) = \mathrm{D}_\sigma(A_\xi^\top y), \tag{5}$$

where the backprojection can be replaced by any pseudoinverse of A_ξ.

Unconditional Generative Networks: Generative models exist in unconditional or conditional forms. Unconditional methods (Bora et al., 2017) (Bora et al., 2018) leverage a pretrained generator $\mathrm{G}_\theta(z) \colon \mathcal{Z} \mapsto \mathcal{X}$ where $z \in \mathcal{Z}$ is a latent code to solve an inverse problem via

$$\mathrm{R}_\theta(y, A_\xi, \sigma) = \mathrm{G}_\theta(\hat{z}) \text{ with } \hat{z} \in \operatorname{argmin}_z f_\sigma(y, A_\xi(\mathrm{G}_\theta(z))) \tag{6}$$

The deep image prior (Ulyanov et al., 2018) is a specific case of unconditional models that uses an untrained generator G_θ, leveraging the inductive bias of a specific autoencoder architecture.

Conditional Generative Networks: Conditional generative adversarial networks (Isola et al., 2017) (Bendel et al., 2023) use adversarial training to learn a network $\mathrm{R}_\theta(y, z, A_\xi, \sigma)$ which provides a set of reconstructions by sampling different latent codes $z \in \mathcal{Z}$.

Foundation Models: Foundation models are end-to-end architectures that incorporate knowledge of A_ξ and σ and are trained to reconstruct images across a wide variety of forward operators A_ξ and noise distributions N_σ (Terris et al., 2025). These models often obtain good performance in new tasks without retraining, and can also be finetuned to specific inverse problems or datasets using measurement data alone.

The table below summarizes the reconstruction methods considered in the library:

Family of Methods	Description	Training	Iterative	Sampling
Artifact Removal	Applies network to a pseudoinverse	Yes	No	No
Variational Optimization	Solves optimization problem with explicit/explicit priors	Hyper-params	Yes	No
Plug-and-Play (PnP)	Leverages pretrained denoisers as priors within an optimization	Hyper-params	Yes	No

DRAFT

Family of Methods	Description	Training	Iterative	Sampling
Unfolded Networks	Trainable architecture by unrolling an optimization	Yes	Only DEQ	No
Diffusion & Langevin	Leverages pretrained denoisers or explicit priors within an ODE/SDE	Hyper-params	Yes	Yes
Generative Adversarial Networks	Model plausible images via generator	Yes	Yes	Depends
Foundation Models	Models trained for many imaging problems	Fine-tune	No	No

Training

Training can be performed using the `Trainer` class, which is a high-level interface for training reconstruction networks. `Trainer` handles data ingestion, the training loop, logging, and checkpointing.

Losses

The library also provides the `loss` module with losses for training R_θ which are specially designed for inverse problems, acting as a framework that unifies loss functions that are widely used in various inverse problems across domains. Loss functions are defined as

$$l = \mathcal{L}\left(\hat{x}, x, y, A_\xi, \mathrm{R}_\theta\right) \tag{7}$$

with $\hat{x} = R_\theta(y, A_\xi, \sigma)$ being the network prediction, and written in deepinv as `l = loss(x_hat, x, y, physics, model)`, where some inputs might be optional (e.g., x is not needed for self-supervised losses).

Supervised Losses: Supervised learning can be done using a dataset of ground-truth and measurements pairs $\{(x_i, y_i)\}_{i=1}^N$ by applying a metric to compute the distance between x and $\hat{x}$. If the forward model is known, measurements are typically generated directly during training from a dataset of ground-truth references $\{x_i\}_{i=1}^N$.

Self-Supervised Losses: Self-supervised losses rely on measurement data only $\{y_i\}_{i=1}^N$. We provide implementations of state-of-the-art losses from the literature (A. Wang & Davies, 2025). These can be roughly classified into three classes:

The first class consists of splitting losses (Batson & Royer, 2019), with operator-specific solutions for denoising (Krull et al., 2019) (Huang et al., 2021) and MRI (Yaman et al., 2020) (Eldeniz et al., 2021) (Liu et al., 2020). The second class is Stein's Unbiased Risk Estimate (SURE) and related losses: we provide variants of SURE for Gaussian, Poisson, and Poisson-Gaussian noise respectively, which can also be used without knowledge of the noise levels (Tachella et al., 2025). The library includes the closely related Recorrupted2Recorrupted (Pang et al., 2021) which handles Gaussian, Poisson, and Gamma noise distributions (Monroy et al., 2025). The third class corresponds to nullspace losses, exploiting invariance of the image distribution to transformations (Chen et al., 2021) or access to multiple forward operators (Tachella et al., 2022).

The library provides image transforms $\tilde{x} = T_g x$ in the `transform` module where $g \in G$ parametrizes a transformation group, including a group-theoretic model for geometric transforms covering linear (Euclidean) and non-linear (projective, diffeomorphism) transforms (A. Wang & Davies, 2024). These can be used for data augmentation, equivariant imaging (Chen et al., 2021), and equivariant networks.

Network Regularization Losses: Network regularization losses which try to enforce some regularity condition on R_θ, such as having an upper bounded Lipschitz constant or similarly being firmly non-expansive (Pesquet et al., 2021).

Adversarial Losses: Adversarial losses are used to train conditional or unconditional generative networks. The library provides both supervised (Bora et al., 2017) and self-supervised (i.e., no ground-truth) (Bora et al., 2018) adversarial losses.

Datasets

The library provides a common framework for defining and simulating datasets for image reconstruction. Datasets return ground-truth and measurements pairs $\{(x_i, y_i)\}_{i=1}^N$, and may also return physics parameters ξ_i. Given a dataset of reference images $\{x_i\}_{i=1}^N$, the library can be used to generate and save a simulated paired dataset to encourage reproducibility. The library also provides interfaces to some popular datasets to facilitate research in specific application domains:

- Natural images: Div2K (Agustsson & Timofte, 2017), Urban100 (Lim et al., 2017), Set14 (Zeyde et al., 2012), CBSD68 (Martin et al., 2001), Flickr2K (Lim et al., 2017), LSDIR (Li et al., 2023);
- MRI scans: FastMRI (Zbontar et al., 2018), CMRxRecon (C. Wang et al., 2024);
- Computed tomography scans: LIDC-IDRI (Armato III et al., 2011);
- Fluorescence microscopy images: FMD (Y. Zhang et al., 2019);
- Real motion blur images: Kohler (Köhler et al., 2012);
- Multispectral satellite images: NBU (Meng et al., 2021).

Evaluation

Reconstruction methods can be evaluated on datasets with the method `Trainer.test` using metrics defined in our framework. These are written in `deepinv` as `m = metric(x_hat, x)` in the case of full-reference metrics, or as `m = metric(x_hat)` for no-reference metrics, and provide common functionality such as input normalization and complex magnitude.

Following the distortion-perception trade-off in image reconstruction problems (Blau & Michaeli, 2018), the library provides common popular distortion metrics such as PSNR, SSIM (Z. Wang et al., 2004), and LPIPS (R. Zhang et al., 2018), as well as no-reference perceptual metrics such as NIQE (Mittal et al., 2012) and QNR (Yeganeh & Wang, 2012), which can be used to evaluate the quality of the reconstructions.

Philosophy

Coding Practices

`deepinv` is coded in modern Python following a test-driven development philosophy. The code is unit-, integration- and performance-tested using `pytest` and verified using `codecov` (automatically via GitHub actions), and is compliant with PEP8 using `black`. To encourage reproducibility, the library passes random number generators for all random generation functionality. Architecturally, `deepinv` is implemented using an object-oriented framework where base classes provide abstract functionality and interfaces (such as `Physics` or `Metric`), sub-classes provide specific implementations or special cases (such as `LinearPhysics`) along

DRAFT

with methods inherited from base classes (such as the operator pseudoinverse), and mixins provide specialized methods. This framework reduces code duplication and makes it easy for researchers, engineers, and practitioners to implement new or specialized functionality while inheriting existing methods.

Documentation

The library provides a **user guide**, quickstart and in-depth **examples** for all levels of user, and individual API documentation for classes. The documentation is built using Sphinx. We use Sphinx-Gallery (Najera et al., 2023) for generating Jupyter notebook demos, which are automatically tested and included in the documentation. The user guide provides an overview of the library and is intended as a starting point for new users, whereas the API lists all classes and functions, which are intended for advanced users. The user guide also serves as a computational imaging tutorial, providing an overview of common imaging operators and reconstruction methods. The documentation of most classes includes a usage example which is automatically tested using `doctest`, and a detailed mathematical description using latex with shared math symbols and notation across the whole documentation.

Perspectives

DeepInverse is a dynamic and evolving project and this paper is merely a snapshot of ongoing progress (release v0.3.0). The community is continuously contributing more methods, such as more realistic physics operators and more advanced training techniques, which reflect the state-of-the-art in imaging with deep learning, aiming to address the needs and interests of researchers and practitioners alike.

Acknowledgements

J. Tachella acknowledges support by the French National Research Agency (Agence Nationale de la Recherche) grant UNLIP (ANR-23-CE23-0013) and the CNRS PNRIA deepinverse project. M. Terris acknowledges support by the BrAIN grant (ANR-20-CHIA-0016). F. Sarron, P. Weiss, M.H. Nguyen were supported by the ANR Micro-Blind ANR-21-CE48-0008. Thomas Moreau was supported from a national grant attributed to the ExaDoST project of the NumPEx PEPR program, under the reference ANR-22-EXNU-0004. J. Hertrich is supported by the German Research Foundation (DFG) with project number 530824055. Z. Hu acknowledges funding from the Swiss National Science Foundation (grant PZ00P2_216211). Thomas Davies is supported by UKRI EPSRC (grants EP/V006134/1 and EP/V006177/1). S. Neumayer acknowledges funding from the German Research Foundation (DFG) with project number 543939932. The authors acknowledge the Jean-Zay high-performance computing center, using HPC resources from GENCI-IDRIS (Grants 2021-AD011012210, 2024-AD011015191).

References

Adler, J., Kohr, H., Ringh, A., Moosmann, J., sbanert, Ehrhardt, M. J., Lee, G. R., niinimaki, bgris, Verdier, O., Karlsson, J., zickert, Palenstijn, W. J., Öktem, O., Chen, C., Loarca, H. A., & Lohmann, M. (2018). *Odlgroup/odl: ODL 0.7.0* (Version v0.7.0). Zenodo. https://doi.org/10.5281/zenodo.1442734

Agustsson, E., & Timofte, R. (2017). NTIRE 2017 challenge on single image super-resolution: Dataset and study. *The IEEE Conference on Computer Vision and Pattern Recognition (CVPR) Workshops*.

Altekrüger, F., Denker, A., Hagemann, P., Hertrich, J., Maass, P., & Steidl, G. (2023). PatchNR: Learning from very few images by patch normalizing flow regularization. *Inverse*

Problems, *39*(6), 064006.

Armato III, S. G., McLennan, G., Bidaut, L., McNitt-Gray, M. F., Meyer, C. R., Reeves, A. P., Zhao, B., Aberle, D. R., Henschke, C. I., Hoffman, E. A., & others. (2011). The lung image database consortium (LIDC) and image database resource initiative (IDRI): A completed reference database of lung nodules on CT scans. *Medical Physics*, *38*(2), 915–931.

Bai, S., Kolter, J. Z., & Koltun, V. (2019). Deep equilibrium models. *Advances in Neural Information Processing Systems*, *32*.

Balke, T., Davis, F., Garcia-Cardona, C., Majee, S., McCann, M., Pfister, L., & Wohlberg, B. (2022). Scientific computational imaging code (SCICO). *Journal of Open Source Software*, *7*(78), 4722. https://doi.org/10.21105/joss.04722

Batson, J., & Royer, L. (2019). Noise2self: Blind denoising by self-supervision. *International Conference on Machine Learning*, 524–533.

Bendel, M., Ahmad, R., & Schniter, P. (2023). A regularized conditional GAN for posterior sampling in image recovery problems. In A. Oh, T. Naumann, A. Globerson, K. Saenko, M. Hardt, & S. Levine (Eds.), *Advances in neural information processing systems* (Vol. 36, pp. 68673–68684). Curran Associates, Inc. https://proceedings.neurips.cc/paper_files/paper/2023/file/d8b29f07599fecdba93d87ed27a65524-Paper-Conference.pdf

Biguri, A., Sadakane, T., Lindroos, R., Liu, Y., Landman, M. S., Du, Y., Lohvithee, M., Kaser, S., Hatamikia, S., Bryll, R., & others. (2025). TIGRE v3: Efficient and easy to use iterative computed tomographic reconstruction toolbox for real datasets. *Engineering Research Express*, *7*(1), 015011.

Blau, Y., & Michaeli, T. (2018). The perception-distortion tradeoff. *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition*, 6228–6237.

Bora, A., Jalal, A., Price, E., & Dimakis, A. G. (2017). Compressed sensing using generative models. *International Conference on Machine Learning*, 537–546.

Bora, A., Price, E., & Dimakis, A. G. (2018). AmbientGAN: Generative models from lossy measurements. *International Conference on Learning Representations*.

Candès, E. J., & Wakin, M. B. (2008). An introduction to compressive sampling. *IEEE Signal Processing Magazine*, *25*(2), 21–30.

Chambolle, A., & Pock, T. (2016). An introduction to continuous optimization for imaging. *Acta Numerica*, *25*, 161–319.

Chen, D., Tachella, J., & Davies, M. E. (2021). Equivariant imaging: Learning beyond the range space. *Proceedings of the IEEE/CVF International Conference on Computer Vision*, 4379–4388.

Chung, H., Kim, J., Kim, S., & Ye, J. C. (2023). Parallel diffusion models of operator and image for blind inverse problems. *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 6059–6069.

Chung, H., Kim, J., Mccann, M. T., Klasky, M. L., & Ye, J. C. (2022). Diffusion posterior sampling for general noisy inverse problems. *arXiv Preprint arXiv:2209.14687*.

Debarnot, V., & Weiss, P. (2024). Deep-blur: Blind identification and deblurring with convolutional neural networks. *Biological Imaging*, *4*, e13.

Dossal, C., Hurault, S., & Papadakis, N. (2024). Optimization with first order algorithms. *arXiv Preprint arXiv:2410.19506*. https://arxiv.org/abs/2410.19506

Eldeniz, C., Gan, W., Chen, S., Fraum, T. J., Ludwig, D. R., Yan, Y., Liu, J., Vahle, T., Krishnamurthy, U., Kamilov, U. S., & others. (2021). Phase2Phase: Respiratory

motion-resolved reconstruction of free-breathing magnetic resonance imaging using deep learning without a ground truth for improved liver imaging. *Investigative Radiology*, *56*(12), 809–819.

Gazzola, S., Hansen, P. C., & Nagy, J. G. (2019). IR tools: A MATLAB package of iterative regularization methods and large-scale test problems. *Numerical Algorithms*, *81*(3), 773–811.

Gossard, A., & Weiss, P. (2024). Training adaptive reconstruction networks for blind inverse problems. *SIAM Journal on Imaging Sciences*, *17*(2), 1314–1346.

Gregor, K., & LeCun, Y. (2010). Learning fast approximations of sparse coding. *Proceedings of the 27th International Conference on International Conference on Machine Learning*, 399–406.

Hestenes, M. R., Stiefel, E., & others. (1952). Methods of conjugate gradients for solving linear systems. *Journal of Research of the National Bureau of Standards*, *49*(6), 409–436.

Ho, J., Jain, A., & Abbeel, P. (2020). Denoising diffusion probabilistic models. *Advances in Neural Information Processing Systems*, *33*, 6840–6851.

Huang, T., Li, S., Jia, X., Lu, H., & Liu, J. (2021). Neighbor2neighbor: Self-supervised denoising from single noisy images. *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 14781–14790.

Isola, P., Zhu, J.-Y., Zhou, T., & Efros, A. A. (2017). Image-to-image translation with conditional adversarial networks. *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition*, 1125–1134.

Jin, K. H., McCann, M. T., Froustey, E., & Unser, M. (2017). Deep convolutional neural network for inverse problems in imaging. *IEEE Transactions on Image Processing*, *26*(9), 4509–4522.

Jørgensen, J. S., Ametova, E., Burca, G., Fardell, G., Papoutsellis, E., Pasca, E., Thielemans, K., Turner, M., Warr, R., Lionheart, W. R., & others. (2021). Core Imaging Library-part I: A versatile python framework for tomographic imaging. *Philosophical Transactions of the Royal Society A*, *379*(2204), 20200192.

Kawar, B., Elad, M., Ermon, S., & Song, J. (2022). Denoising diffusion restoration models. *Advances in Neural Information Processing Systems*, *35*, 23593–23606.

Köhler, R., Hirsch, M., Mohler, B., Schölkopf, B., & Harmeling, S. (2012). Recording and playback of camera shake: Benchmarking blind deconvolution with a real-world database. *Computer Vision–ECCV 2012: 12th European Conference on Computer Vision, Florence, Italy, October 7-13, 2012, Proceedings, Part VII 12*, 27–40.

Kowalski, M. (2009). Sparse regression using mixed norms. *Applied and Computational Harmonic Analysis*, *27*(3), 303–324.

Krull, A., Buchholz, T.-O., & Jug, F. (2019). Noise2void-learning denoising from single noisy images. *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2129–2137.

Laumont, R., Bortoli, V. D., Almansa, A., Delon, J., Durmus, A., & Pereyra, M. (2022). Bayesian imaging using plug & play priors: When Langevin meets Tweedie. *SIAM Journal on Imaging Sciences*, *15*(2), 701–737.

Lazarus, C., Weiss, P., Chauffert, N., Mauconduit, F., El Gueddari, L., Destrieux, C., Zemmoura, I., Vignaud, A., & Ciuciu, P. (2019). SPARKLING: Variable-density k-space filling curves for accelerated T2*-weighted MRI. *Magnetic Resonance in Medicine*, *81*(6), 3643–3661.

Li, Y., Zhang, K., Liang, J., Cao, J., Liu, C., Gong, R., Zhang, Y., Tang, H., Liu, Y., Demandolx, D., & others. (2023). Lsdir: A large scale dataset for image restoration.

Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, 1775–1787.

Lim, B., Son, S., Kim, H., Nah, S., & Mu Lee, K. (2017). Enhanced deep residual networks for single image super-resolution. *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition Workshops*, 136–144.

Liu, J., Sun, Y., Eldeniz, C., Gan, W., An, H., & Kamilov, U. S. (2020). RARE: Image reconstruction using deep priors learned without groundtruth. *IEEE Journal of Selected Topics in Signal Processing*, *14*(6), 1088–1099.

Mallat, S. (1999). *A wavelet tour of signal processing*. Elsevier.

Martin, D., Fowlkes, C., Tal, D., & Malik, J. (2001). A database of human segmented natural images and its application to evaluating segmentation algorithms and measuring ecological statistics. *Proc. 8th Int'l Conf. Computer Vision*, *2*, 416–423.

Meng, X., Xiong, Y., Shao, F., Shen, H., Sun, W., Yang, G., Yuan, Q., Fu, R., & Zhang, H. (2021). A large-scale benchmark data set for evaluating pansharpening performance: Overview and implementation. *IEEE Geoscience and Remote Sensing Magazine*, *9*(1), 18–52. https://doi.org/10.1109/MGRS.2020.2976696

Mittal, A., Soundararajan, R., & Bovik, A. C. (2012). Making a "completely blind" image quality analyzer. *IEEE Signal Processing Letters*, *20*(3), 209–212.

Monroy, B., Bacca, J., & Tachella, J. (2025). Generalized recorrupted-to-recorrupted: Self-supervised learning beyond gaussian noise. *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*.

Najera, O., Larson, E., Liu, L., Esteve, L., Varoquaux, G., Grobler, J., Andrade, E. S. de, Holdgraf, C., Gramfort, A., Jas, M., Nothman, J., Rehberg, S., Grisel, O., Varoquaux, N., Hiscocks, S., alexis, Gouillart, E., Hoffmann, T., Lee, A., … Kunzmann, P. (2023). *Sphinx-gallery/sphinx-gallery: v0.12.2* (Version v0.12.2). Zenodo. https://doi.org/10.5281/zenodo.7716999

Nehme, E., Freedman, D., Gordon, R., Ferdman, B., Weiss, L. E., Alalouf, O., Naor, T., Orange, R., Michaeli, T., & Shechtman, Y. (2020). DeepSTORM3D: Dense 3D localization microscopy and PSF design by deep learning. *Nature Methods*, *17*(7), 734–740.

Ong, F., & Lustig, M. (2019). *SigPy: A python package for high performance iterative reconstruction*. ISMRM. https://archive.ismrm.org/2019/4819.html

Paige, C. C., & Saunders, M. A. (1975). Solution of sparse indefinite systems of linear equations. *SIAM Journal on Numerical Analysis*, *12*(4), 617–629.

Paige, C. C., & Saunders, M. A. (1982). LSQR: An algorithm for sparse linear equations and sparse least squares. *ACM Transactions on Mathematical Software (TOMS)*, *8*(1), 43–71.

Pang, T., Zheng, H., Quan, Y., & Ji, H. (2021). Recorrupted-to-recorrupted: Unsupervised deep learning for image denoising. *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2043–2052.

Paszke, A., Gross, S., Massa, F., Lerer, A., Bradbury, J., Chanan, G., Killeen, T., Lin, Z., Gimelshein, N., Antiga, L., & others. (2019). PyTorch: An imperative style, high-performance deep learning library. *Advances in Neural Information Processing Systems*, *32*.

Pereyra, M., Mieles, L. V., & Zygalakis, K. C. (2020). Accelerating proximal markov chain monte carlo by using an explicit stabilized method. *SIAM Journal on Imaging Sciences*, *13*(2), 905–935. https://doi.org/10.1137/19M1283719

Pesquet, J.-C., Repetti, A., Terris, M., & Wiaux, Y. (2021). Learning maximally monotone operators for image recovery. *SIAM Journal on Imaging Sciences*, *14*(3), 1206–1237.

Polson, L. A., Fedrigo, R., Li, C., Sabouri, M., Dzikunu, O., Ahamed, S., Karakatsanis, N., Kurkowska, S., Sheikhzadeh, P., Esquinas, P., & others. (2025). Pytomography: A python library for medical image reconstruction. *SoftwareX*, *29*, 102020.

Ravasi, M., & Vasconcelos, I. (2019). PyLops–a linear-operator python library for large scale optimization. *arXiv Preprint arXiv:1907.12349*.

Riis, N. A., Alghamdi, A. M., Uribe, F., Christensen, S. L., Afkham, B. M., Hansen, P. C., & Jørgensen, J. S. (2024). CUQIpy: I. Computational uncertainty quantification for inverse problems in python. *Inverse Problems*, *40*(4), 045009.

Romano, Y., Elad, M., & Milanfar, P. (2017). The little engine that could: Regularization by denoising (RED). *SIAM Journal on Imaging Sciences*, *10*(4), 1804–1844.

Rudin, L. I., Osher, S., & Fatemi, E. (1992). Nonlinear total variation based noise removal algorithms. *Physica D: Nonlinear Phenomena*, *60*(1-4), 259–268.

Simeoni, M., Kashani, S., Rué-Queralt, J., & Developers, P. (2024). *Pyxu-org/pyxu: pyxu*. Zenodo. https://doi.org/10.5281/zenodo.4486431

Song, Y., Sohl-Dickstein, J., Kingma, D. P., Kumar, A., Ermon, S., & Poole, B. (2020). Score-based generative modeling through stochastic differential equations. *arXiv Preprint arXiv:2011.13456*.

Soubies, E., Soulez, F., McCann, M. T., Pham, T., Donati, L., Debarre, T., Sage, D., & Unser, M. (2019). Pocket guide to solve inverse problems with GlobalBioIm. *Inverse Problems*, *35*(10), 104006.

Tachella, J., Chen, D., & Davies, M. (2022). Unsupervised learning from incomplete measurements for inverse problems. *Advances in Neural Information Processing Systems*, *35*, 4983–4995.

Tachella, J., Davies, M., & Jacques, L. (2025). UNSURE: Self-supervised learning with unknown noise level and stein's unbiased risk estimate. *International Conference on Learning Representations*.

Terris, M., Hurault, S., Song, M., & Tachella, J. (2025). Reconstruct anything model: A lightweight foundation model for computational imaging. *arXiv Preprint arXiv:2503.08915*.

Terris, M., & Moreau, T. (2023). Meta-prior: Meta learning for adaptive inverse problem solvers. *arXiv Preprint arXiv:2311.18710*.

Ulyanov, D., Vedaldi, A., & Lempitsky, V. (2018). Deep image prior. *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition*, 9446–9454.

Van Aarle, W., Palenstijn, W. J., Cant, J., Janssens, E., Bleichrodt, F., Dabravolski, A., De Beenhouwer, J., Joost Batenburg, K., & Sijbers, J. (2016). Fast and flexible x-ray tomography using the ASTRA toolbox. *Optics Express*, *24*(22), 25129–25147.

Van der Vorst, H. A. (1992). Bi-CGSTAB: A fast and smoothly converging variant of bi-CG for the solution of nonsymmetric linear systems. *SIAM Journal on Scientific and Statistical Computing*, *13*(2), 631–644.

Venkatakrishnan, S. V., Bouman, C. A., & Wohlberg, B. (2013). Plug-and-play priors for model based reconstruction. *2013 IEEE Global Conference on Signal and Information Processing*, 945–948.

Wang, A., & Davies, M. (2024). Perspective-equivariance for unsupervised imaging with camera geometry. *IEEE/CVF European Conference on Computer Vision (ECCV) Workshop on Traditional Computer Vision in the Age of Deep Learning*.

Wang, A., & Davies, M. (2025). Benchmarking self-supervised methods for accelerated MRI reconstruction. *arXiv Preprint arXiv:2502.14009*. https://arxiv.org/abs/2502.14009

DRAFT

Wang, C., Lyu, J., Wang, S., Qin, C., Guo, K., Zhang, X., Yu, X., Li, Y., Wang, F., Jin, J., & others. (2024). CMRxRecon: A publicly available k-space dataset and benchmark to advance deep learning for cardiac MRI. *Scientific Data*, *11*(1), 687.

Wang, Z., Bovik, A. C., Sheikh, H. R., & Simoncelli, E. P. (2004). Image quality assessment: From error visibility to structural similarity. *IEEE Transactions on Image Processing*, *13*(4), 600–612.

Yaman, B., Hosseini, S. A. H., Moeller, S., Ellermann, J., Uğurbil, K., & Akçakaya, M. (2020). Self-supervised learning of physics-guided reconstruction neural networks without fully sampled reference data. *Magnetic Resonance in Medicine*, *84*(6), 3172–3191.

Yeganeh, H., & Wang, Z. (2012). Objective quality assessment of tone-mapped images. *IEEE Transactions on Image Processing*, *22*(2), 657–667.

Zbontar, J., Knoll, F., Sriram, A., Murrell, T., Huang, Z., Muckley, M. J., Defazio, A., Stern, R., Johnson, P., Bruno, M., & others. (2018). FastMRI: An open dataset and benchmarks for accelerated MRI. *arXiv Preprint arXiv:1811.08839*.

Zeyde, R., Elad, M., & Protter, M. (2012). On single image scale-up using sparse-representations. *Curves and Surfaces: 7th International Conference, Avignon, France, June 24-30, 2010, Revised Selected Papers 7*, 711–730.

Zhang, K., Li, Y., Zuo, W., Zhang, L., Van Gool, L., & Timofte, R. (2021). Plug-and-play image restoration with deep denoiser prior. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, *44*(10), 6360–6376.

Zhang, K., Zuo, W., Chen, Y., Meng, D., & Zhang, L. (2017). Beyond a gaussian denoiser: Residual learning of deep cnn for image denoising. *IEEE Transactions on Image Processing*, *26*(7), 3142–3155.

Zhang, R., Isola, P., Efros, A. A., Shechtman, E., & Wang, O. (2018). The unreasonable effectiveness of deep features as a perceptual metric. *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition*, 586–595.

Zhang, Y., Zhu, Y., Nichols, E., Wang, Q., Zhang, S., Smith, C., & Howard, S. (2019). A poisson-gaussian denoising dataset with real fluorescence microscopy images. *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 11710–11718.

Zhu, Y., Zhang, K., Liang, J., Cao, J., Wen, B., Timofte, R., & Van Gool, L. (2023). Denoising diffusion models for plug-and-play image restoration. *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 1219–1229.

Zoran, D., & Weiss, Y. (2011a). From learning models of natural image patches to whole image restoration. *2011 International Conference on Computer Vision*, 479–486.

Zoran, D., & Weiss, Y. (2011b). From learning models of natural image patches to whole image restoration. *2011 International Conference on Computer Vision*, 479–486. https://doi.org/10.1109/ICCV.2011.6126278

DRAFT